STUDENT WORKSHEET

Bridge Design — Two Evidence Options

IRVING Independent School District | CAREER & COLLEGE EXPLORATIONS

NAME ______________________ | DATE ______________________ | PERIOD ________

Constraints and Evidence Boundary

- 12-inch clear span between secured supports
- teacher-provided prepacked material limit
- one marked load point
- same known staged loads and stop rule for every physical test
- equal fixed-data route when the physical test gate is not met

This classroom prototype does not validate a real bridge. Real work requires licensed analysis, codes, materials testing, geotechnical evidence, public review, and longer validation.

Bridge Systems

- **Beam:** deck/member spans supports; performance depends on geometry, material, connections, span, and loading.
- **Truss:** connected members form triangular units that can stabilize an idealized pin-jointed frame; joint/member behavior still matters.
- **Arch:** curved form transfers compression toward supports/abutments; support behavior matters.

Word bank: span · support · member · joint · load point · load path · weak point

Design question: How will the complete system carry the marked load toward the supports?

Option A — Top and Side Views

Label 12-inch span, supports, members, joints, and marked load point.

Top view

Side view

Expected load path and predicted weak point:

Option B — Top and Side Views

Label 12-inch span, supports, members, joints, and marked load point.

Top view

Side view

Expected load path and predicted weak point:

Critique, Choice, and Career Role

Evidence-based strength of Option A:

Evidence-based strength of Option B:

Chosen option and reason:

Career/role responsible for the next step, evidence needed, and work product:

Use this frame: I selected Option [A/B] because [evidence]. Next, a [role] would use [evidence] to produce [work product].